

Deep Reinforcement Learning

Model-based reinforcement learning

Prof. Dr. Sebastian Peitz

Chair of Safe Autonomous Systems, TU Dortmund

Summer term 2026

Content

Content

- Revisiting the past: Dyna-style algorithms
- Looking into the future: Planning at decision time
 - MCTS
 - AlphaZero
- Learning a model
 - Dreamer architecture
 - Uncertainty estimation?
 - Learned ensembles

Where are we?

Chapter	Topic	Content
	Basics & tabular methods	
1-5	Bandits, MDPs, Dynamic Programming, Monte Carlo, TD Learning	RL basics in finite dimensions
	Deep-learning-based methods	
6-13	DQN, policy gradients, actor-critic, PPO/SAC, exploration	Deep RL from basics to modern algorithms
	Model-Based Control	
14	Optimal control & feedback control	How to do control when we know the model
15	Model-based reinforcement learning	RL with known and learned models
	Advanced Topics	

Lecture contents

Revisiting the past: Dyna-style algorithms

Looking into the future: Planning at decision time

Example: AlphaZero

Learning a model

TBD

References